



Cloudflare Radar 2024 Year in Review

Global Internet traffic patterns, adoption trends,
connectivity disruptions, and security insights

17% Global Traffic Growth and Starlink's 3.3x Surge

17%

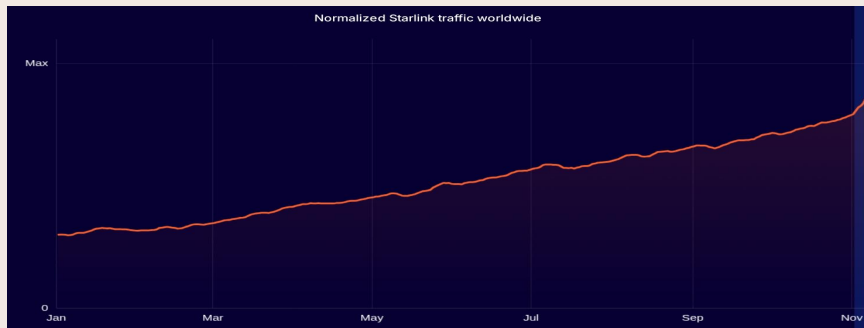
Year-Over-Year Traffic Growth

Global Internet traffic continues upward trajectory driven by communication, commerce, entertainment, and transportation

3.3x

Starlink Traffic Growth

Satellite connectivity expands to underserved regions worldwide



Key Insight

Starlink's remarkable 3.3x traffic surge reflects the growing demand for satellite-based Internet connectivity in regions with limited infrastructure.

Mobile devices account for 41% of global traffic.

Google Leads Top 10 Services as Generative AI Rises

Rank	Internet Service
1	Google
2	Facebook
3	Apple
4	TikTok
5	AWS
6	Microsoft
7	Instagram
8	YouTube
9	Amazon
10	WhatsApp

Based on aggregate data from Cloudflare's 1.1.1.1 DNS resolver

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
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Generative AI

Continued meteoric rise throughout 2024, while the Metaverse receded into the background

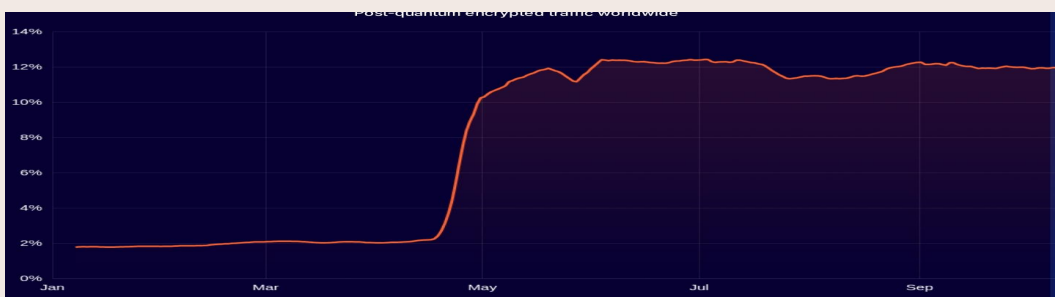
Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

2% → 13%

Post-Quantum TLS 1.3 Traffic

Adoption surge following Chrome 124 default enablement (April 2024)

● Early 2024: ~2% Post-Chrome 124: ~13% ●



21%

HTTP/3 Traffic

Go

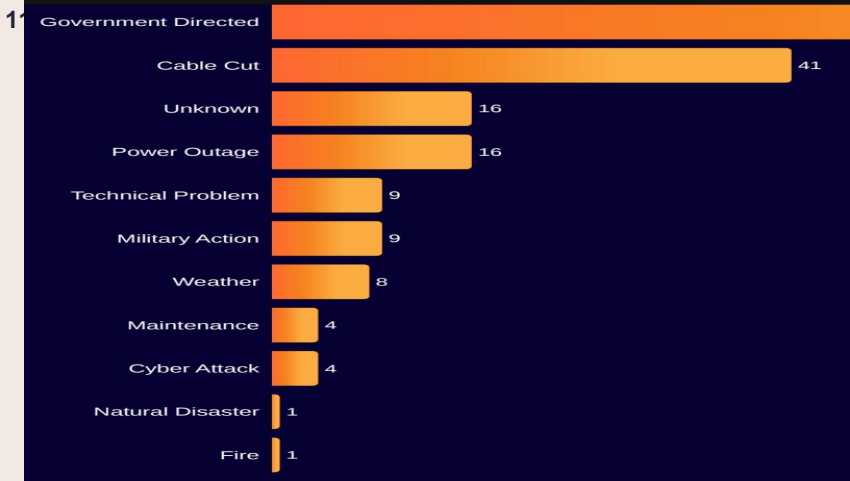
Top API Client Language (~12%)

API Client Languages

Go: ~12%
Node.js: (next)
Python: 9.6%
Java: 7.4%
.NET: 3.6%

225 Major Disruptions: Government Shutdowns Lead Outage Causes

Government Directed	
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1



Leading Cause: Government-Directed Shutdowns

116+ incidents driven by various factors including political unrest, censorship requirements, and national security concerns.

Cable cuts (41) and power outages (16) follow as second and third leading causes.

225

Major Internet Disruptions Observed in 2024

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

6.5%

Traffic Mitigated

Most Attacked

Gambling & Games

Industry Sector

Bot Traffic

68%

from Top 10 Countries

Log4j Exploitation Persists

CVE-2021-44228

First disclosed December 2021 — still actively targeted 2+ years later, underscoring persistent nature of widely known vulnerabilities.

Email Security

4.3% of emails processed were malicious

43% of malicious emails contained deceptive links

bar TLD originated largest share of malicious and spam email

RPKI Valid Routes

+6.4%

An important step toward improved Internet routing security

Five Takeaways Shaping the 2025 Internet Outlook

01

Traffic Growth Trajectory

17% YoY growth continues; Starlink's 3.3x surge demonstrates satellite connectivity expansion to underserved regions

02

Post-Quantum Encryption Momentum

From ~2% to 13% of TLS 1.3 traffic following Chrome 124 default enablement — expect continued acceleration

03

Connectivity Fragility

225 major disruptions observed; government-directed shutdowns (116+) remain the leading cause of Internet outages

04

Persistent Security Threats

6.5% traffic mitigated; Gambling/Games most attacked; Log4j exploitation continues years after disclosure

05

Evolving Developer Ecosystem

Go surpasses other languages as top API client (~12%); HTTP/3 at 21% adoption; mobile drives 41% of traffic